

AIM: Write a program to find precision, accuracy, f1, recall, false positive, false negative from two 1-D dataset one is the predicted value another one is the original value also plot the confusion matrix of the dataset

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

print("Enter Actual values :")
actual = list(map(int, input().split()))

print("Enter Predicted values :")
predicted = list(map(int, input().split()))

cm = confusion_matrix(actual, predicted)
tn, fp, fn, tp = cm.ravel()

total = tp + tn + fp + fn
accuracy = (tp + tn) / total
precision = tp / (tp + fp)
recall = tp / (tp + fn)
f1 = 2 * (precision * recall) / (precision + recall)

print("True Positives (TP):", tp)
print("True Negatives (TN):", tn)
print("False Positives (FP):", fp)
print("False Negatives (FN):", fn)

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)

print("F1 Score:", f1)

disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix")
plt.show()
```

INPUT:

PROBLEMS OUTPUT DEBUG CONSOLE

```
Enter Actual values :  
1 0 1 0 1 1  
Enter Predicted values :  
0 1 1 1 0 1
```

OUTPUT:

```
True Positives (TP): 2  
True Negatives (TN): 0  
False Positives (FP): 2  
False Negatives (FN): 2  
Accuracy: 0.3333333333333333  
Precision: 0.5  
Recall: 0.5  
F1 Score: 0.5
```

PLOT:

